Lovera Lokeswara

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Education

University of Waterloo, Canada *Sep 2025 – Dec 2026*
Master of Data Science and Artificial Intelligence (Co-op Program), **Grade: 96**
Waterloo AI Club Machine Learning Engineer: FlockRL (Reinforcement Learning Drones)

The Hong Kong University of Science and Technology, Hong Kong SAR *Sep 2021 – Dec 2024*
BSc in Data Science and Technology, **CGPA: 3.7 (First Class Honors)**, 2022 & 2024 Dean's List Scholar

Work Experience

Shopify *Canada*
Machine Learning Engineer Intern *May 2026 – Aug 2026*

- Built a deduplication system for Shopify's AI-powered knowledge management pipeline, preventing 50% of duplicate fixes by detecting semantically similar documentation changes using embeddings, document overlap analysis, and LLM-based classification (86.4% accuracy).
- Diagnosed and resolved production issues blocking automated Help Center documentation updates, including corrupted GitHub API operations, reducing failed operation time from 330s to 30s and restoring reliable automated deployments.

Bridge AI Knowledge *Remote*
Software Engineer *Mar 2025 – May 2025*

- Designed and deployed a PDF Processing API (Python) with integrated LLM chatbot and automated parser to extract information from 100+ page documents, reducing processing time by 80% and enabling 5x higher document throughput.
- Built automated compliance report generator that benchmarks documents against client-specific checklists and provides recommendations which resulted in 95% audit accuracy and reducing report generation time by 95% (2 hours to 5 minutes).
- Developed WhatsApp customer service agent by integrating A1Base with N8N workflows, automating 100% of routine product inquiries and reducing average response time from 5 minute to 10 seconds while decreasing manual support workload.

Cathay Pacific Airways *Hong Kong SAR*
Data Scientist Intern *Jul 2024 – Aug 2024*

- Developed a Bayesian machine learning model using PyMC to analyze 55,000 user reports, enabling the engineering team to identify and prioritize key areas for issue resolution in Cathay's in-flight entertainment system.
- Created a solution to improve in-flight crew service effectiveness through smart wearable technology that leverages AI and machine learning algorithms with the goal of ensuring seamless interactions between cabin crews and customers.
- Acted as Project Manager for Cathay Hackathon 2024. Organized and led initiatives by adopting an agile approach with external partners (i.e., Huawei, Apple, Microsoft, Alibaba, Amazon) progressing the event significantly towards completion.

TradeLink Technologies *Hong Kong SAR*
Data Analyst, Business Intelligence *Jun 2023 – Aug 2023*

- Built a feature that allowed clients to access relevant news pertaining to their shipment status. Feature was developed through SEO practices which filters for news centered on a client.
- Researched and analyzed internal data to identify discrepancies in shipment and order records to ensure exact documentation of data within our client facing application.

Projects

Additional technical projects are archived in detail and can be accessed through: github.com/LoveraLokeswara

Optimization of a Gromov–Wasserstein Solver (Research)

- Formulated and analyzed the Gromov–Wasserstein optimal transport problem for low-dimensional point clouds, focusing on optimization strategies that guarantee globally optimal solutions.
- Optimized a Gromov–Wasserstein solver in Python (POT/OT, Gurobi) and evaluated global convergence on point-cloud benchmarks, achieving faster convergence and fewer iterations than public baselines.

Financial Volatility Forecasting

- Developed a comprehensive Python/Jupyter framework to forecast financial time-series volatility by comparing traditional econometric models (ARIMA, GARCH) with machine learning approaches (LSTM, SVM) across assets such as S&P 500 and EUR/USD.
- Evaluated hybrid models (e.g., ARIMA-LSTM, GARCH-LSTM) to capture both linear dynamics and nonlinear patterns, improving forecast robustness and offering quantitative insights for risk management applications.

Financial Sentiment Analysis using NLP

- Built a financial sentiment analysis pipeline using transformer-based models (FinBERT, BERT, RoBERTa) to classify financial text into positive, negative, or neutral sentiments, enabling nuanced market sentiment insights for investment and risk analysis.

Technical Skills

Technical Skills: Python, C/C++, Git, R, MLflow, Power BI, SQL, MATLAB, Oracle, UI/UX, VBA, VS Code/Visual Studio, Econometrics, Regression Analysis, Statistical Modelling, Mathematical Modelling, Apache Spark